distance. Nevertheless, we can state that the force is becoming already very small at a distance of 5×10^{-13} cm. This means that the force between proton and neutron has an extremely short range, and in this respect it is very different from an electric force.

Instead of studying the force, we may base our considerations on the potential energy which a neutron has within the field of a proton (or vice versa). When the distance is a very large one, we arbitrarily assign to this energy the magnitude 0. At finite distances, it has a negative magnitude. Due to the limited range, the potential energy is practically 0 for any but a very small distance. Figure 15 shows an approximate curve of the potential energy as a function of the distance r. This energy increases rapidly from high negative magnitudes until it reaches the vicinity of 0, which it then approaches asymptotically. Dealing with small distances, the potential energy curve can be computed indirectly from the mass defect of the deuteron. In addition to its potential energy, the system has also a kinetic energy, since the proton and neutron reciprocally vibrate with reference to each other, with a continuous transformation of kinetic energy into potential energy, and vice versa. The sum total of these energies always equals the binding energy, 2·2 Mev., which is shown in Figure 15 as a horizontal line. The average magnitude of the kinetic energy can be estimated, for instance, in accordance with the uncertainty principle, from the diameter of the deuteron. The accuracy of our knowledge of the exact location of this object of finite diameter goes hand in hand with a proportionate inaccuracy of our knowledge of its velocity; and the squared velocity times one-half the mass gives us the magnitude of the average kinetic energy. Once we know both the kinetic energy and the total energy, we can calculate from them the potential energy. Figure 15 is the result of considerations of this nature.

Short-range forces are known to exist in nature elsewhere, too; the most important example is exhibited by the chemical forces, the so-called *valencies* (unless we are dealing with what is called a *polar* combination), in other words, the forces, which, for instance, bind two hydrogen atoms to one oxygen atom in a water molecule. These, too, are short-range forces which are operative actually only when the atoms are in direct contact, but become infinitesimally small as soon as the distance between the atoms increases.

It is due to this extremely short range that in heavier, macroscopic structures we can perceive neither the chemical nor the nuclear forces, whereas electric or magnetic forces are perceptible without any difficulty. The force between two magnetic poles is felt directly by the hand in which a magnet is held, and your hair stands on end as you approach an electrical high-tension apparatus. But chemical forces can never be perceived in this direct fashion, for they are operative over molecular distances only. The same applies to nuclear forces; they cannot be perceived anywhere except in the